

$$\underset{\sim}{Y} \sim N(\underset{\sim}{X}\underset{\sim}{\beta}, \sigma^2 I_n)$$

$$\underset{\sim}{Y} \in \mathbb{R}^n \quad \underset{\sim}{\varepsilon} \in \mathbb{R}^n$$

$$\underset{\sim}{X} \in \mathbb{R}^{n \times (k+1)}$$

$$\underset{\sim}{\beta} \in \mathbb{R}^{k+1}, \quad \sigma^2 > 0$$

Hence we have $(k+2)$ unknown model parameters $\underset{\sim}{\beta}, \sigma$.

$$\downarrow$$

$$(\beta_0, \beta_1, \dots, \beta_k)$$

Problem Statement :-

Estimation :-

If $\hat{\underset{\sim}{\beta}}$ minimizes the least square condition then it satisfies the normal eqⁿ.

$$2S(\underset{\sim}{\beta}) /$$

$\frac{\partial \text{SSE}}{\partial \beta}$

$$* \hat{\beta}_{\sim} = (X^T X)^{-1} X^T Y$$

$$* \hat{y}_{\sim} = X \hat{\beta}_{\sim} = X (X^T X)^{-1} X^T Y = P_X Y \rightarrow \text{Projection of } y \text{ in } C(X).$$

$$* e_{\sim} = Y - \hat{y}_{\sim} = (I - P_X) Y$$

$$* e_{\sim}^T \hat{y}_{\sim} = 0.$$

ANOVA :-

↳ Analysis of variance

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_K = 0$$

H_1 : H_0 is not true

Under H_0 : $y_i \stackrel{iid}{\sim} N(\beta_0, \sigma^2)$
 $\bar{y}$ estimated as $\bar{y}$.

$$* SS_{Total} = \sum_{i=1}^n (y_i - \bar{y})^2$$

$$= \underset{\sim}{y}^T \left(I_n - \frac{1}{n} \underset{\sim}{1} \underset{\sim}{1}^T \right) \underset{\sim}{y}$$

$$* SS_{Error} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = (\underset{\sim}{y} - \underset{\sim}{\hat{y}})^T (\underset{\sim}{y} - \underset{\sim}{\hat{y}})$$

$$= \left[(I_n - P_x) \underset{\sim}{y} \right]^T \left[(I_n - P_x) \underset{\sim}{y} \right]$$

$$= \underset{\sim}{y}^T (I_n - P_x) \underset{\sim}{y}$$

$$* SS_{Model} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \underset{\sim}{y}^T \left(P_x - \frac{1}{n} \underset{\sim}{1} \underset{\sim}{1}^T \right) \underset{\sim}{y}$$

$$* SS_{Total} = \underset{\sim}{y}^T \left(I_n - \frac{1}{n} \underset{\sim}{1} \underset{\sim}{1}^T \right) \underset{\sim}{y}$$

$$= \underset{\sim}{y}^T \left(I_n - P_x + P_x - \frac{1}{n} \underset{\sim}{1} \underset{\sim}{1}^T \right) \underset{\sim}{y}$$

$$\begin{aligned}
&= \underset{\sim}{Y}^T (I_n - P_X) \underset{\sim}{Y} + \underset{\sim}{Y}^T \left(P_X - \frac{1}{n} \underset{\sim}{1} \underset{\sim}{1}^T \right) \underset{\sim}{Y} \\
&= \text{SS Error} + \text{SS Model}
\end{aligned}$$

* Distribution of SS :-

$$\frac{\text{SS Error}}{\sigma^2} \sim \chi^2 \left(\begin{array}{l} \text{dof} = n - k - 1 \\ \text{nep} = 0 \end{array} \right) \quad \begin{array}{l} \nearrow \text{Rank}(I_n - P_X) \\ \text{chi squared distrib.} \end{array}$$

$$\frac{\text{SS Model}}{\sigma^2} \sim \chi^2 \left(\begin{array}{l} \text{dof} = k \\ \text{nep} = \lambda > 0 \end{array} \right)$$

$$\frac{\text{SS Total}}{\sigma^2} \sim \chi^2 \left(\begin{array}{l} \text{dof} = n - 1 \\ \text{nep} = \lambda > 0 \end{array} \right)$$

* We will show that under H_0 , $\lambda = 0$ and under H_1 , $\lambda > 0$.

* If $\lambda > 0$, $\left(\frac{\text{SS Model}}{\sigma^2} \right) / k \sim F$

$$\left(\frac{SS_{Error}}{\sigma^2} \right) / (n-k-1) \quad \text{disbri}$$

$$\left(\frac{SS_{Error}}{\sigma^2} \right) / (n-k-1)$$

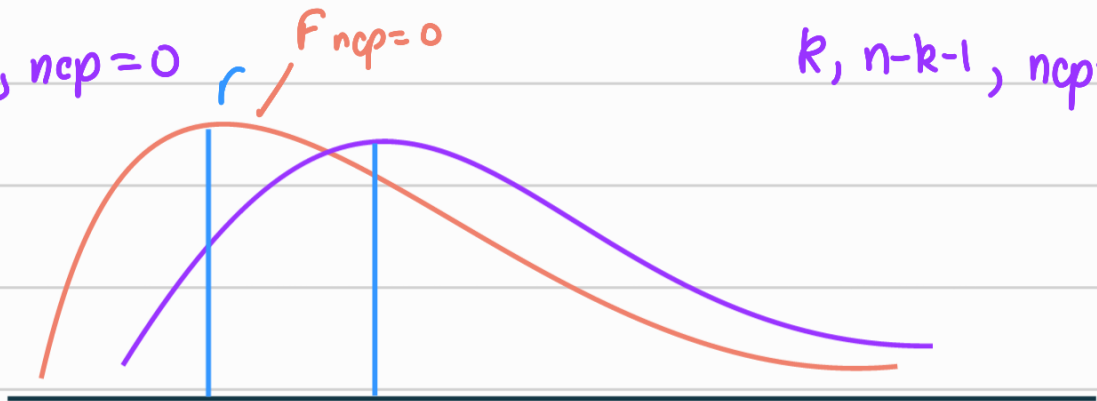


Under H_0

$k, n-k-1, ncp=0$

Under H_1

$k, n-k-1, ncp=\lambda > 0$



we will reject H_0 in favour of H_1 if $F_{obs} > F_{\alpha, k, n-k-1}$



6th Feb, 2023

$$Y \sim N(X\beta, I_n \sigma^2)$$

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

$$\hat{\beta} \sim N(\beta, (X^T X)^{-1} \sigma^2)$$

* ANOVA Hypothesis :-

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$$

H_1 : H_0 is not true

Alternatively,

$$\begin{aligned} H_0 : \beta_j &= b_j \\ H_1 : \beta_j &\neq b_j \end{aligned} \quad \left. \begin{array}{l} \text{If } b_j = 0, x_j \text{ won't have} \\ \text{and contribution to the model.} \end{array} \right\}$$

Alternatively,

$$\begin{aligned} H_0 : \beta_i - \theta \beta_j &= b \quad (=0) \\ H_1 : \beta_i - \theta \beta_j &\neq b \quad (\neq 0) \end{aligned} \quad \left. \begin{array}{l} \text{Features } x_1 \text{ and} \\ x_2 \text{ are linearly} \\ \text{dependent.} \end{array} \right\}$$

Alternatively,

$$H_0 : \beta_i = \beta_j = 0$$

H_1 : H_0 is not true.

Testing the Hypothesis :-

$$\textcircled{\text{ii}} \quad H_0 : \beta_j = b_j$$

$$H_1 : \beta_j \neq b_j$$

$$(X^T X)^{-1} = C$$

$$(n \times 1) (1 \times n) \rightarrow C_{ij}$$

↳ starts from 0-k

$$l_j^T \hat{\beta} = \hat{\beta}_j \sim N(l_j^T \beta, \sigma^2 l_j^T C l_j)$$

$$= N(\beta_j, \sigma^2 C_{jj})$$

$$l_j = \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}$$

$$C_j = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ 0 \end{pmatrix} \rightarrow j^{\text{th}} \text{ element}$$

$$\hat{\beta}_j - \beta_j \sim N(0, \sigma^2 C_{jj})$$

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\sigma^2 C_{jj}}} \sim N(0, 1)$$

$\rightarrow \sigma^2$ is unknown, hence we will use an estimator.

$$\hat{\sigma}^2 = \frac{SSE_{\text{error}}}{n-k-1} = MS_{\text{Error}} = \frac{Y^T (I_n - P_X) Y}{n-k-1}$$

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} = \frac{(\hat{\beta}_j - \beta_j) / \sqrt{\sigma^2 C_{jj}}}{\sqrt{\left(\frac{Y^T (I_n - P_X) Y}{\sigma^2} \right) / (n-k-1)}} \quad \left. \begin{array}{l} \text{Normally} \\ \text{distributed} \end{array} \right\}$$

$\rightarrow$ Chi-squared distributed.

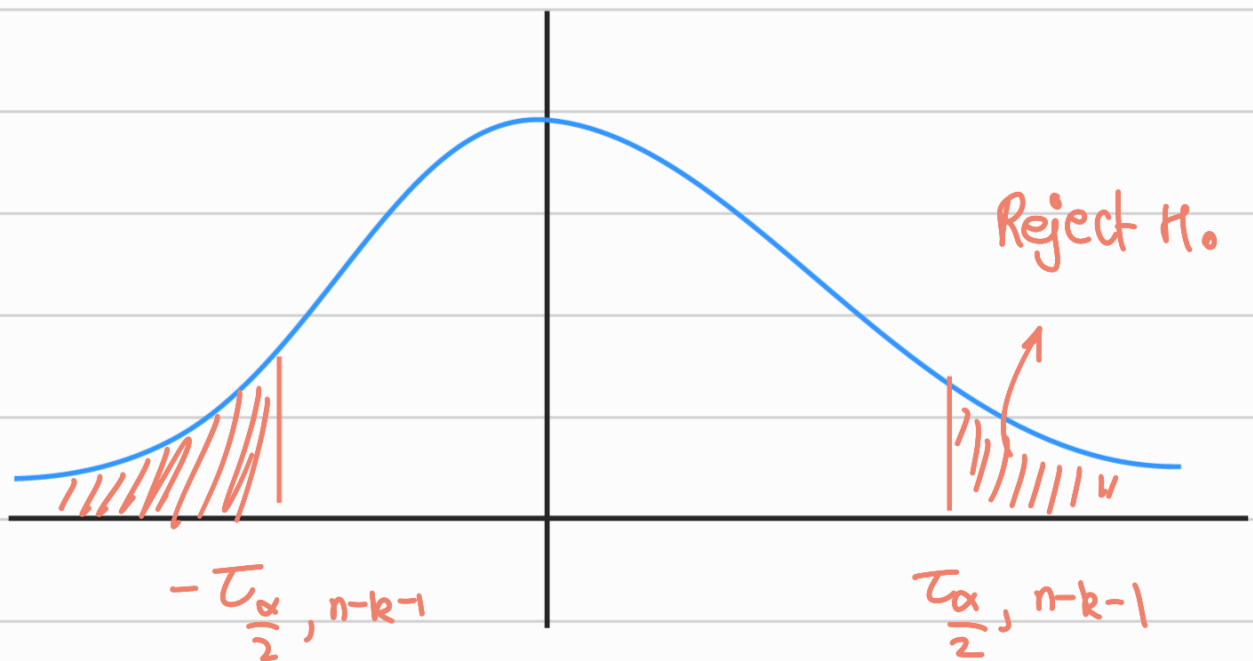
Numerator from
projection & denominator
from error

Both numerator and
denominator are independently
distributed

$$\sim t_{n-k-1}$$

Under H_0 ,

$$\frac{\hat{\beta}_j - b_j}{\sqrt{\hat{\sigma}^2 C_{jj}}} \sim t_{n-k-1}$$



(iii) $H_0 : \beta_i - \theta \beta_j = 0$
 $H_1 : \beta_i - \theta \beta_j > 0$

$$\underline{l} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ -\theta \end{pmatrix} \begin{matrix} \rightarrow i \\ \rightarrow j \end{matrix}$$

$$\beta_i - \theta \beta_j = \underline{l}^T \underline{\beta}$$

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\hat{\beta} \sim N(\beta, \sigma^2 C^{-1})$$

$$= N(\beta_i - \theta \beta_j, \sigma^2 (c_{ii} - \theta c_{ij} - \theta c_{ji} + \theta^2 c_{jj}))$$

$$= N(\beta_i - \theta \beta_j, \sigma^2 (c_{ii} - 2\theta c_{ij} + \theta^2 c_{jj}))$$

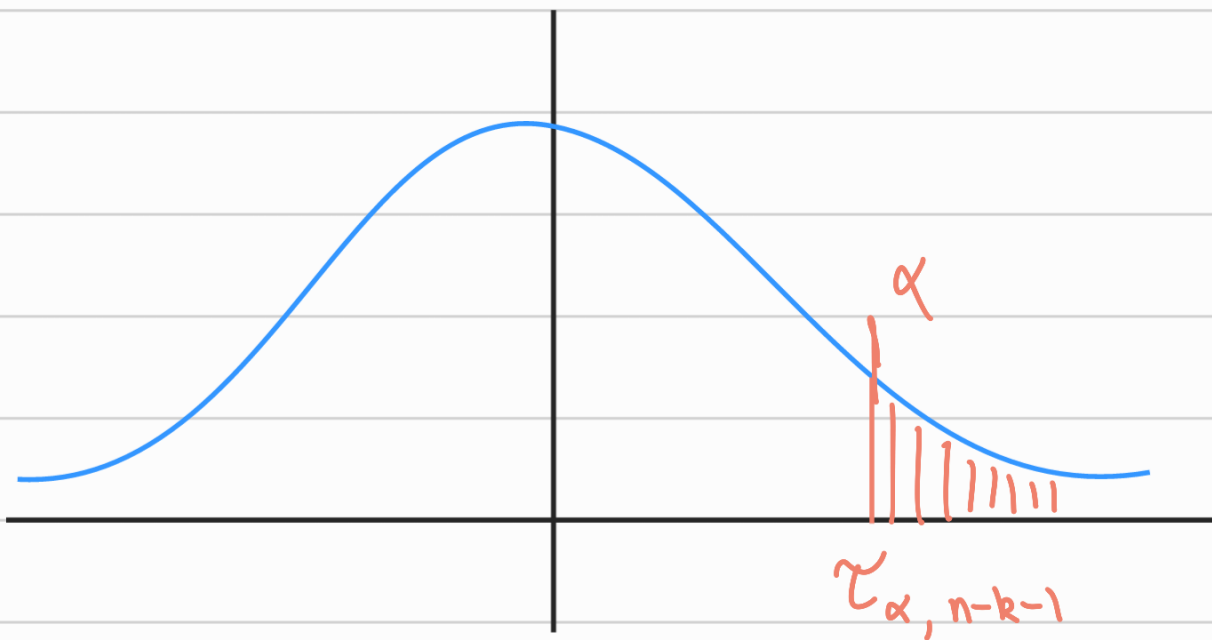
$$\hat{\sigma}^2 = \frac{SS_{Error}}{n-k-1} = MS_{Error} = \frac{Y^T (I_n - P_X) Y}{n-k-1}$$

$$\frac{\hat{\beta} - \beta}{\sqrt{\hat{\sigma}^2 C^{-1}}} \sim t_{n-k-1}$$

$$\text{Under } H_0, \quad T = \frac{\hat{\beta} - b}{\sqrt{\hat{\sigma}^2 C^{-1}}} \sim t_{n-k-1}$$

We reject H_0 in favour of H_1 at level α if

$$T_{obs} > T_{\alpha, n-k-1} \iff P_{value} < \alpha$$



(N)

$$H_0 : \begin{cases} \beta_i = 0 \\ \beta_j = 0 \end{cases}$$

$H_1 : H_0$ is not true

OR

$$H_0 : \begin{cases} \beta_i = 0 \\ \beta_i - \beta_j = 0 \end{cases}$$

$H_1 : \text{—————}$

Here we have two hypothesis in H_0 and we have to construct a method to test both of them simultaneously.

$$A = \begin{bmatrix} 0 & \dots & \overset{i^{\text{th}}}{\downarrow} 1 & 0 & \dots & 0 \\ 0 & \dots & 0 & \underset{\uparrow}{1} & \dots & 0 \end{bmatrix}$$

jth

$$A\beta = 0$$

$$\hat{\beta} \sim N(\beta, \sigma^2)$$

$$A\hat{\beta} \sim N(A\beta, \sigma^2 ACA^T)$$

$$(A\hat{\beta} - A\beta) \sim N(0, \sigma^2 ACA^T)$$

$$(A\hat{\beta} - A\beta)^T (\sigma^2 ACA^T)^{-1} (A\hat{\beta} - A\beta) \sim \chi^2_{\text{dof} = \text{rank}(A)} \\ \text{ncp} = 0$$

$$(A\hat{\beta} - A\beta)^T (\hat{\sigma}^2 ACA^T)^{-1} (A\hat{\beta} - A\beta) \sim \chi^2_{\text{dof} = \text{rank}(A)}$$

7th Feb, 2023

— x — x — x —

Best Estimator:

A linear function is BLUE of its expectation

iff it is uncorrelated with all linear zero functions.
(LZF)

Corollary: If L_y^T is an LUE of $P^T \beta$ then
 $L^T P_x y$ is the BLUE of $P^T \beta$

→ As σ^2 is unknown,

$$\hat{\sigma}^2 = \text{MSE}_{\text{Error}} = \frac{\text{SSE}_{\text{Error}}}{n - (k+1)}$$

$$\frac{(A\hat{B} - AB)^T (ACA^T)^{-1} (A\hat{B} - AB) / \text{rank}(A)}{(y^T (I_n - P_x) y) / (n - k - 1)} \sim F_{\substack{df_1 = \text{rank}(A) \\ df_2 = n - k - 1}}$$

Under H_0 ,

$$\frac{(A\hat{B} - \underline{b})^T (ACA^T)^{-1} (A\hat{B} - \underline{b}) / \text{rank}(A)}{(y^T (I_n - P_x) y) / (n - k - 1)} \sim F_{\substack{\text{rank}(A) \\ n - k - 1}}$$

↓
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upper tail test.

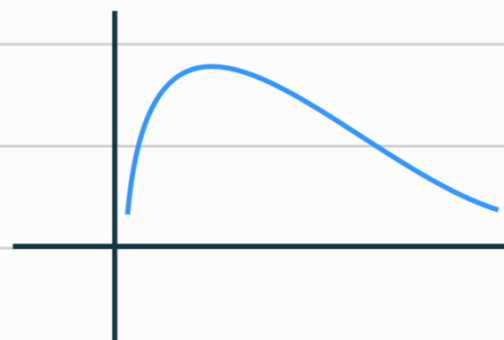
Here, we will use a T^2 -test,
which is an upper tail test.

$$X \sim N(\mu, \sigma^2)$$

$$H_0: \mu = \mu_0 \mid H_1: \mu \neq \mu_0$$

$$\textcircled{*} T = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\hat{\sigma}} \sim t_{n-1}$$

$$\textcircled{*} T^2 \sim F_{1, n-1} \text{ (upper tail)}$$



ANOVA Hypothesis

$$H_0: \beta_1 = \beta_2 = \dots = \beta_k = 0$$

H_1 : H_0 is not true

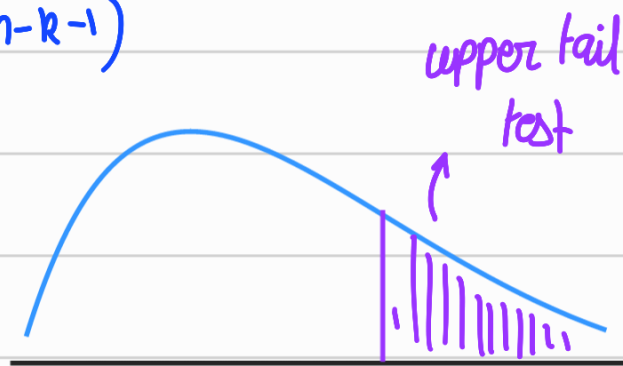
$$A\underset{\sim}{\beta} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \underset{\sim}{\beta} = \underset{\sim}{0}$$

$k \times (k+1)$

$$F = (\hat{A}\hat{\beta})^T (\hat{A}CA^T)^{-1} (\hat{A}\hat{\beta}) / k$$

$$\frac{(\tilde{y}^T (I_n - P_X) \tilde{y})}{(n-k-1)}$$

$$F \stackrel{H_0}{\sim} F_{k, n-k-1}$$



$$F = \frac{\left(\frac{SS \text{ Model}}{\sigma^2} \right) / k}{\left(\frac{SS \text{ Error}}{\sigma^2} \right) / (n-k-1)}$$

$$\begin{array}{ccc} SS_{\text{Total}} & = & SS_{\text{Model}} + SS_{\text{Error}} \\ \begin{array}{c} (n-1) \\ ncp=\lambda \end{array} & & \begin{array}{c} (k) \\ ncp=\lambda \end{array} \quad \begin{array}{c} (n-k-1) \\ ncp=0 \end{array} \end{array}$$

$$\tilde{y} \sim N(\tilde{X}\beta, \sigma^2 I_n) \Rightarrow \frac{\tilde{y}}{\sigma} \sim N\left(\frac{\tilde{X}\beta}{\sigma}, I_n\right)$$

$$\frac{SS \text{ Model}}{\sigma^2} = \left(\frac{\tilde{X}\beta}{\sigma} \right)^T \left(P_X - \frac{1}{n} \tilde{1} \tilde{1}^T \right) \left(\frac{\tilde{X}\beta}{\sigma} \right)$$

$$= \frac{1}{\sigma^2} \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_R \end{pmatrix} \begin{bmatrix} 0 & 0' \\ 0 & X_R^T X_R - \frac{1}{n} X_R \mathbf{1} \mathbf{1}^T X_R \end{bmatrix} \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_R \end{pmatrix}$$

$$\text{Cov}(\underline{\tilde{X}}) = \frac{1}{n-1} \left[X_R^T X_R - \frac{1}{n} X_R^T \mathbf{1} \mathbf{1}^T X_R \right]$$

For $n_{cp} = 0$, $\beta_R = 0$

Coefficient of Determination -

$$R^2 = \frac{\text{Variation in } Y \text{ explained by the model}}{\text{Total variation in } Y}$$

$$= \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = \frac{\text{SS Model}}{\text{SS Total}}$$

$$= 1 - \frac{\text{SS Error}}{\text{SS Total}}$$

$$= 1 - \frac{\sum e_i^2}{\text{SS Total}}$$

$$\sum (y_i - \bar{y})^2$$

$$R^2_{adj} = 1 - \frac{\sum_{i=1}^n e_i^2 / df}{\sum_{i=1}^n (y_i - \bar{y})^2 / df}$$

$$= 1 - \frac{\sum_{i=1}^n e_i^2 / (n-k-1)}{\sum (y_i - \bar{y})^2 / (n-1)} < R^2$$

* R^2 is an increasing function while $adj(R^2)$ is not.

Linear Unbiased Estimator (LUE)

A linear estimator $C^T y = \sum l_i y_i$ is said to be a linear unbiased estimator (LUE) of $p^T \beta$ if,

$$E(C^T y) = p^T \beta \quad (\forall \beta \in \mathbb{R}^{k+1})$$

Result-1

The iff condition of a LPF $\underset{\sim}{p}^T \underset{\sim}{\beta}$ to be estimable is $\underset{\sim}{p} \in C(X^T)$

Linear Parametric function

By defi,

$$E\left(\underset{\sim}{l}^T \underset{\sim}{y}\right) = \underset{\sim}{p}^T \underset{\sim}{\beta} \quad \forall \underset{\sim}{\beta} \in \mathbb{R}^{k+1}$$

$$\underset{\sim}{l}^T X \underset{\sim}{\beta} = \underset{\sim}{p}^T \underset{\sim}{\beta} \quad \forall \underset{\sim}{\beta} \in \mathbb{R}^{k+1}$$

$$\underset{\sim}{l}^T X = \underset{\sim}{p}^T \iff X^T \underset{\sim}{l} = \underset{\sim}{p} \Rightarrow \underset{\sim}{p} \in C(X^T)$$

Result-2

Show that $\underset{\sim}{l}^T y$ is a LZF iff $X^T \underset{\sim}{l} = 0$.

by defi,

$$E(\underset{\sim}{l}^T y) = 0 \quad \forall \underset{\sim}{\beta} \in \mathbb{R}^{k+1}$$

$$\underset{\sim}{l}^T X \underset{\sim}{\beta} = \underset{\sim}{0}^T \underset{\sim}{\beta} = 0 \quad \forall \underset{\sim}{\beta} \in \mathbb{R}^{k+1}$$

$$\Rightarrow \underset{\sim}{l}^T X = 0^T$$

$$\Rightarrow \underset{\sim}{l} \in (C(X))^{\perp}$$

$$\Rightarrow \underset{\sim}{l} \in C(I_n - P_X)$$

If $\underset{\sim}{l} \in C(I_n - P_X)$ then there exist some $\underset{\sim}{m}$ such that,

$$\underset{\sim}{l} = (I_n - P_X) \underset{\sim}{m}$$

Now, we can see that,

$$\begin{aligned} l^T X &= \left[(I_n - P_x) \tilde{m} \right]^T X \\ &= \tilde{m}^T \left[(I_n - P_x) \right] X = \tilde{m}^T \tilde{0} = 0 \end{aligned}$$

∴ Every LZF will have the form, $\tilde{m}^T (I_n - P_x) Y$.
L Linear Zero Functions

⑧ Let $\tilde{l}^T \tilde{y}$ is a LUE of LPF $\tilde{p}^T \tilde{\beta}$,
 ie, $E(\tilde{l}^T \tilde{y}) = \tilde{p}^T \tilde{\beta} \quad \forall \tilde{\beta}$.

Then the BLUE of $\tilde{p}^T \tilde{\beta}$ is $\tilde{l}^T P_x \tilde{y}$.

$$\begin{aligned} \textcircled{1} \quad \tilde{l}^T \tilde{y} &= \tilde{l}^T I_n \tilde{y} = \tilde{l}^T (P_x + I_n - P_x) \tilde{y} \\ &= \tilde{l}^T P_x \tilde{y} + \tilde{l}^T (I_n - P_x) \tilde{y} \end{aligned}$$

$$\textcircled{2} \quad E(\tilde{l}^T P_x \tilde{y}) = \tilde{l}^T P_x X \beta = (\tilde{l}^T X) \beta = \tilde{p}^T \tilde{\beta} \text{ is a LUE.}$$

$$\textcircled{3} \quad E(\tilde{l}^T (I_n - P_x) \tilde{y}) = 0 \text{ is a LZF}$$

$$\begin{aligned} \textcircled{4} \quad \text{cov}(\tilde{l}_x^T P_x \tilde{y}, \tilde{l}_x^T (I_n - P_x) \tilde{y}) &= \tilde{l}_x^T P_x (\sigma^2 I_n) (I_n - P_x) \\ &= \sigma^2 \tilde{l}_x^T \cdot P_x (I_n - P_x) \tilde{l}_x = 0 \end{aligned}$$

Hence $\tilde{l}_x^T P_x \tilde{y}$ is BLUE of $\tilde{p}^T \tilde{\beta}$.

$\textcircled{5}$ BLUE is unique.

↓

Let if possible $\tilde{l}_1^T \tilde{y}$ and $\tilde{l}_2^T \tilde{y}$ both are BLUEs of $\tilde{p}^T \tilde{\beta}$.

$$\begin{aligned} \tilde{l}_1^T \tilde{y} &= \tilde{l}_2^T \tilde{y} + \tilde{l}_1^T \tilde{y} - \tilde{l}_2^T \tilde{y} \\ &= \tilde{l}_2^T \tilde{y} + (\tilde{l}_1 - \tilde{l}_2)^T \tilde{y} \end{aligned}$$

$$E(\tilde{l}_2^T \tilde{y}) = \tilde{p}^T \tilde{\beta}$$

$$E((\tilde{l}_1 - \tilde{l}_2)^T \tilde{y}) = 0 \Rightarrow \text{cov}(\tilde{l}_2^T \tilde{y}, (\tilde{l}_1 - \tilde{l}_2)^T \tilde{y}) = 0$$

$$* \text{var}(\tilde{l}_1^T \tilde{y}) = \text{var}(\tilde{l}_2^T \tilde{y}) + \text{var}(\tilde{l}_1^T - \tilde{l}_2^T) \tilde{y}$$

$$\text{if } \text{var}((\tilde{l}_1^T - \tilde{l}_2^T) \tilde{y}) \neq 0 \rightarrow \text{var}(\tilde{l}_1^T \tilde{y}) > \text{var}(\tilde{l}_2^T \tilde{y})$$

contradiction.

$$\text{if } \text{var}((l_1^T - l_2^T) \tilde{y}) = 0 \rightarrow E((l_1 - l_2)^T \tilde{y}) = 0$$

$$\Rightarrow P((l_1 - l_2)^T \tilde{y} = 0) = 1$$



$$\circ \circ \quad P(l_1^T \tilde{y} = l_2^T \tilde{y}) = 1.$$

